Symmetric graph matrices and spectral clustering

MATH 2740 – Mathematics of Data Science – Unit U10

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The University of Manitoba campuses are located on original lands of Anishinaabeg, Ininew, Anisininew, Dakota and Dene peoples, and on the National Homeland of the Red River Métis. We respect the Treaties that were made on these territories, we acknowledge the harms and mistakes of the past, and we dedicate ourselves to move forward in partnership with Indigenous communities in a spirit of Reconciliation and collaboration.

Outline

- Undirected graphs and similarity graphs
- The combinatorial graph Laplacian
- Spectral properties of the Laplacian
- Graph cuts, the Fiedler vector and spectral clustering

From directed rankings to undirected communities

In Units 08 and 09, we studied directed graphs, non-negative matrices and the Perron-Frobenius theorem

There, the dominant eigenvector $\boldsymbol{v}_1 > \mathbf{0}$ associated with $\rho(A)$ answered a *ranking* question: which vertices are most central or visited most often?

In this unit, we turn to **undirected graphs** and **similarity networks**, where the associated matrices are *real and symmetric*

The main question shifts from ranking individual vertices to **clustering**: how do we partition a graph (or a dataset) into cohesive communities with minimal connections between them?

Why real symmetric matrices change everything

When a matrix $M \in \mathcal{M}_n(\mathbb{R})$ is symmetric ($M^T = M$), the spectral theorem gives us three powerful tools that general directed graphs lack:

1. Every eigenvalue $\lambda_1, \ldots, \lambda_n$ of M is **real**

2. Eigenvectors from distinct eigenspaces are **mutually orthogonal**, and $\mathbb{R}^n$ has an **orthonormal basis of eigenvectors** $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$

3. Eigenvalues and eigenvectors are characterised variationally through the **quadratic form** $\boldsymbol{x}^T M \boldsymbol{x}$ and the **Rayleigh quotient** $\dfrac{\boldsymbol{x}^T M \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}}$

Roadmap of Unit 10

1. **Undirected graphs and similarity graphs:** turning point clouds $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^p$ into symmetric adjacency and weight matrices

2. **The combinatorial graph Laplacian $L = D - A$:** degree matrix, incidence factorisation $L = B^T B$ and the quadratic form $\boldsymbol{x}^T L \boldsymbol{x} = \sum_{\{i,j\} \in E} a_{ij}(x_i - x_j)^2$

3. **Spectral properties of L:** positive semi-definiteness, $\lambda_1 = 0$ with eigenvector $\mathbb{1}$ and why the multiplicity of $\lambda = 0$ counts connected components

4. **Graph cuts, the Fiedler vector and spectral clustering:** expressing $\mathrm{cut}(S, \bar{S}) = \frac{1}{4} \boldsymbol{x}^T L \boldsymbol{x}$, spectral relaxation and partitioning with the Fiedler vector $\boldsymbol{v}_2$

Undirected graphs and similarity graphs

The combinatorial graph Laplacian

Spectral properties of the Laplacian

Graph cuts, the Fiedler vector and spectral clustering

Undirected simple graphs

Let $G = (V, E)$ be an undirected simple graph with vertex set $V = \{1, 2, \ldots, n\}$ and edge set E consisting of unordered pairs $\{i, j\}$ of distinct vertices

- Undirected means $\{i, j\} = \{j, i\}$: if i is connected to j, then j is connected to i
- Simple means there are no self-loops ($\{i, i\} \notin E$) and at most one edge between any pair of vertices

Definition 155 (Adjacency matrix of an undirected graph)

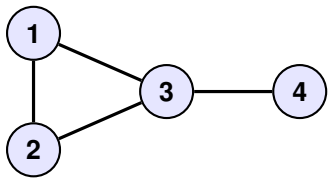
*The **adjacency matrix** of an undirected simple graph $G = (V, E)$ on n vertices is the matrix $A = [a_{ij}] \in \mathcal{M}_n(\{0, 1\})$ defined by*

$$a_{ij} = \begin{cases} 1 & \text{if } \{i, j\} \in E \\ 0 & \text{otherwise} \end{cases}$$

Symmetry of the adjacency matrix

Because $\{i,j\} \in E \iff \{j,i\} \in E$ and $\{i,i\} \notin E$, the adjacency matrix A of an undirected simple graph always satisfies

$$A^T = A \quad (a_{ij} = a_{ji} \ \ \forall i,j) \qquad \text{and} \qquad a_{ii} = 0 \quad (\forall i = 1, \ldots, n)$$



$$A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = A^T$$

Notice that the trace of A is $\text{tr}(A) = \sum_{i=1}^{n} a_{ii} = 0$, so the n real eigenvalues of A always sum to 0 (some are positive and some are negative)

Weighted similarity graphs

In data science, edges often carry a nonnegative weight measuring the **similarity** (or affinity) between vertices i and j

Definition 156 (Weighted similarity graph and weight matrix)

A **weighted undirected graph** $G = (V, E, W)$ has a symmetric weight matrix $W = [w_{ij}] \in \mathcal{M}_n(\mathbb{R}_{\geq 0})$ satisfying

$$w_{ij} = w_{ji} \geq 0 \quad (\forall i, j = 1, \ldots, n) \qquad \text{and} \qquad w_{ii} = 0 \quad (\forall i = 1, \ldots, n)$$

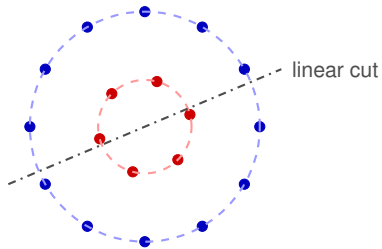
Here $w_{ij} > 0$ if $\{i, j\} \in E$ and $w_{ij} = 0$ if $\{i, j\} \notin E$

- Large w_{ij} means vertices i and j are **strongly connected** (highly similar)
- $w_{ij} = 0$ (or $w_{ij} \approx 0$) means vertices i and j are **unconnected** (dissimilar)
- An unweighted graph is simply the special case $W = A \in \mathcal{M}_n(\{0, 1\})$

Why turn a dataset into a similarity graph?

Suppose we are given n observations $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n \in \mathbb{R}^p$ and want to partition them into clusters

- Centroid-based methods (k-means) and linear projections (PCA) rely on **global Euclidean distances** $\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2$ and linear separating hyperplanes
- They work well when clusters are compact, convex clouds
- They fail completely when clusters have **non-convex geometry** (such as concentric rings, spirals or interlocking crescents), where points in the same cluster are far apart in $\mathbb{R}^p$ but connected through a chain of nearby neighbours



Representing the data as a **local similarity graph** turns non-convex geometric clustering into a **graph connectivity** problem

Constructing similarity graphs from data points

Given observations $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^p$ with pairwise distances $d(\boldsymbol{x}_i, \boldsymbol{x}_j) = \|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2$, there are three standard ways to build a symmetric similarity graph:

1. **ε-neighbourhood graph**: Fix a threshold $\varepsilon > 0$ and connect $i \neq j$ with an unweighted edge ($a_{ij} = 1$) if and only if

$$\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2 \leq \varepsilon$$

2. ***k*-nearest neighbour (*k*-NN) graph**: Connect i and j if $\boldsymbol{x}_i$ is among the k closest points to $\boldsymbol{x}_j$ *or* $\boldsymbol{x}_j$ is among the k closest points to $\boldsymbol{x}_i$ (symmetrised so $A^T = A$)
3. **Gaussian (RBF) similarity graph**: Connect all pairs $i \neq j$ with weight

$$w_{ij} = \exp\left(-\frac{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2}{2\sigma^2}\right), \qquad w_{ii} = 0$$

where $\sigma > 0$ is the bandwidth (scale) parameter

Example: from 4 points in $\mathbb{R}^2$ to a similarity graph

Consider $n = 4$ points in $\mathbb{R}^2$: two nearby points $\boldsymbol{x}_1 = (0,0)^T, \boldsymbol{x}_2 = (0,1)^T$ on the left and two nearby points $\boldsymbol{x}_3 = (3,0)^T, \boldsymbol{x}_4 = (3,1)^T$ on the right

Pairwise Euclidean distances $\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2$:

$$\Delta = \begin{pmatrix} 0 & 1 & 3 & \sqrt{10} \\ 1 & 0 & \sqrt{10} & 3 \\ 3 & \sqrt{10} & 0 & 1 \\ \sqrt{10} & 3 & 1 & 0 \end{pmatrix}$$

With $\varepsilon = 1.5$, the **ε-neighbourhood graph** has adjacency matrix

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

With $\sigma = 1$, the **Gaussian similarity matrix** $w_{ij} = e^{-\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2/2}$ $(i \neq j)$ is

$$W \approx \begin{pmatrix} 0 & 0.607 & 0.011 & 0.007 \\ 0.607 & 0 & 0.007 & 0.011 \\ 0.011 & 0.007 & 0 & 0.607 \\ 0.007 & 0.011 & 0.607 & 0 \end{pmatrix}$$

- Within-cluster weights: $e^{-1/2} \approx 0.607$
- Between-cluster weights: $e^{-9/2} \approx 0.011$ and $e^{-10/2} \approx 0.007$

Role of the scale parameter ε, k or σ

In all three constructions, the scale parameter controls which pairs are treated as local neighbours:

Scale too small



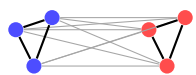
True clusters shatter into many isolated pieces

Well-chosen scale



Dense within clusters, sparse or zero between clusters

Scale too large



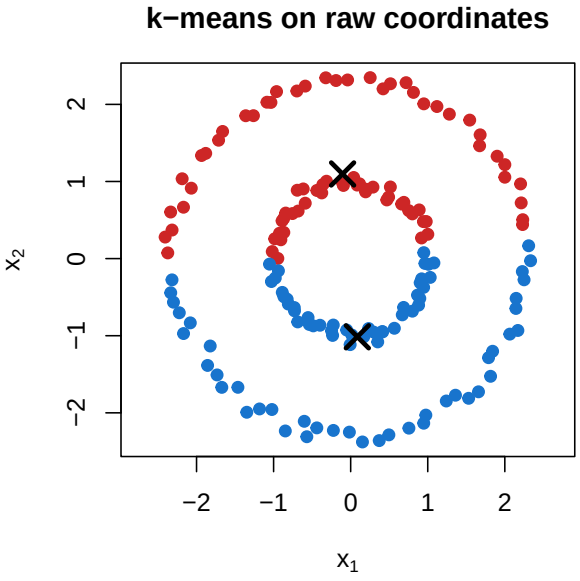
Shortcut edges merge distinct clusters together

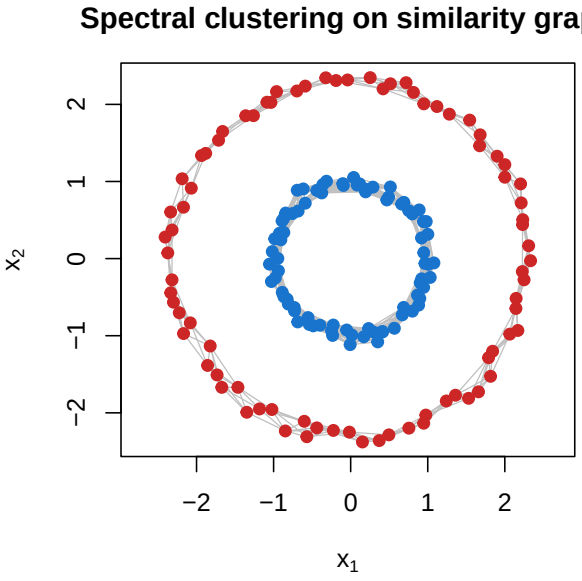
In what follows, we write $A = [a_{ij}]$ for the adjacency or similarity matrix (whether $a_{ij} \in \{0, 1\}$ or $a_{ij} = w_{ij} \geq 0$): all formulas hold identically in both cases

R illustration: two concentric rings

Let us generate $n = 160$ points on two concentric circles in $\mathbb{R}^2$ and compare k-means in $\mathbb{R}^2$ against an ε-neighbourhood graph

```
set.seed(2740)
n1 = 80; n2 = 80
theta1 = seq(0, 2 * pi, length.out = n1 + 1)[-(n1 + 1)]
theta2 = seq(0, 2 * pi, length.out = n2 + 1)[-(n2 + 1)]
X = rbind(
  cbind(1.0 * cos(theta1), 1.0 * sin(theta1)) + rnorm(2 * n1, sd = 0.06),
  cbind(2.3 * cos(theta2), 2.3 * sin(theta2)) + rnorm(2 * n2, sd = 0.06)
)
# epsilon-neighbourhood adjacency matrix (epsilon = 0.65)
dist_mat = as.matrix(dist(X))
A_rings = (dist_mat <= 0.65 & dist_mat > 0) * 1
```


k−means on raw coordinates
x2
x1
2
1
0
−1
−2
−2
−1
0
1
2


Spectral clustering on similarity graph
x2
x1
2
1
0
−1
−2
−2
−1
0
1
2

Undirected graphs and similarity graphs

The combinatorial graph Laplacian

Spectral properties of the Laplacian

Graph cuts, the Fiedler vector and spectral clustering

The combinatorial graph Laplacian

Definition and basic examples

The Laplacian quadratic form

Vertex degrees and the degree matrix

Definition 157 (Vertex degree and degree matrix)

Let $G = (V, E)$ be an undirected graph on n vertices with symmetric adjacency (or weight) matrix $A = [a_{ij}] \in \mathcal{M}_n(\mathbb{R}_{\geq 0})$. The ***degree*** *of vertex i is the row sum*

$$d_i = \sum_{j=1}^{n} a_{ij} = (A\mathbb{1})_i \qquad (i = 1, \ldots, n)$$

where $\mathbb{1} = (1, 1, \ldots, 1)^T \in \mathbb{R}^n$. The ***degree matrix*** *of G is the diagonal matrix*

$$D = \operatorname{diag}(d_1, d_2, \ldots, d_n) = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix}$$

The combinatorial graph Laplacian

Definition 158 (Combinatorial graph Laplacian)

Let $G = (V, E)$ be an undirected graph on n vertices with degree matrix D and adjacency (or weight) matrix A. The ***combinatorial graph Laplacian*** *(or simply* ***Laplacian matrix****) of G is the $n \times n$ matrix*

$$L = D - A$$

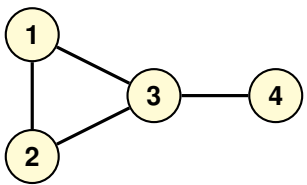
Entry-by-entry, $L = [\ell_{ij}]$ is given by

$$\ell_{ij} = \begin{cases} d_i & \text{if } i = j \\ -a_{ij} & \text{if } i \neq j \end{cases}$$

For a simple unweighted graph, $\ell_{ii} = d_i$, $\ell_{ij} = -1$ if $\{i, j\} \in E$ and $\ell_{ij} = 0$ otherwise

Example 1: Laplacian of a 4-vertex paw graph

Consider the 4-vertex graph $G = (V, E)$ below (a triangle $\{1, 2, 3\}$ with a pendant edge $\{3, 4\}$):



Degrees: $d_1 = 2$, $d_2 = 2$, $d_3 = 3$, $d_4 = 1$

$$D = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

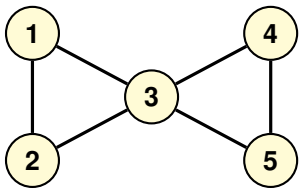
$$L = D - A = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

Quick sanity check on L:

- L is symmetric ($L^T = L$), with positive degrees on the diagonal and -1 on edges
- Every row and every column of L sums to 0: e.g. row 3 has $(-1) + (-1) + 3 + (-1) = 0$

Example 2: Laplacian of a 5-vertex bowtie graph

Consider the 5-vertex bowtie graph formed by two triangles $\{1, 2, 3\}$ and $\{3, 4, 5\}$ sharing central vertex 3:



Degrees: $d_1 = d_2 = d_4 = d_5 = 2$, $d_3 = 4$

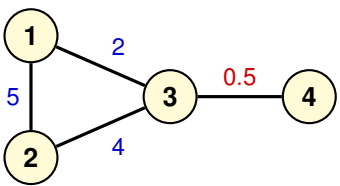
$$L = D - A = \begin{pmatrix} 2 & -1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ -1 & -1 & 4 & -1 & -1 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & -1 & 2 \end{pmatrix}$$

Notice how the structure of L reflects the graph topology:

- The upper-left 2×2 block $\{1, 2\}$ and lower-right 2×2 block $\{4, 5\}$ have no direct edges between them (0 in the off-diagonal blocks)
- All paths between $\{1, 2\}$ and $\{4, 5\}$ pass through vertex 3 (row and column 3)

Example 3: Laplacian of a weighted similarity graph

For a weighted graph $G = (V, E, W)$, the exact same recipe $L = D - W$ applies using weighted degrees $d_i = \sum_{j=1}^{n} w_{ij}$:



Weighted degrees:
$d_1 = 5 + 2 = 7,\ d_2 = 5 + 4 = 9$
$d_3 = 2 + 4 + 0.5 = 6.5,\ d_4 = 0.5$

$$W = \begin{pmatrix} 0 & 5 & 2 & 0 \\ 5 & 0 & 4 & 0 \\ 2 & 4 & 0 & 0.5 \\ 0 & 0 & 0.5 & 0 \end{pmatrix}$$

$$L = D - W = \begin{pmatrix} 7 & -5 & -2 & 0 \\ -5 & 9 & -4 & 0 \\ -2 & -4 & 6.5 & -0.5 \\ 0 & 0 & -0.5 & 0.5 \end{pmatrix}$$

Every row and column of L still sums to 0, and
$\mathrm{tr}(L) = \sum_{i=1}^{n} d_i = 2 \sum_{\{i,j\} \in E} w_{ij} = 23$

Connection with the oriented incidence matrix

We can also factor L as a Gram matrix $B^T B$ by assigning an arbitrary orientation to each edge of $G = (V, E)$

Definition 159 (Oriented edge-vertex incidence matrix)

Let $G = (V, E)$ have $n = |V|$ vertices and $m = |E|$ edges $e_1, \ldots, e_m$. Choose an arbitrary direction $e_k = (i, j)$ for each edge $(i \to j)$. The **oriented incidence matrix** $B = [b_{kv}] \in \mathcal{M}_{mn}(\{-1, 0, 1\})$ has one row per edge and one column per vertex:

$$b_{kv} = \begin{cases} +1 & \text{if } v = i \text{ (tail of } e_k) \\ -1 & \text{if } v = j \text{ (head of } e_k) \\ 0 & \text{otherwise} \end{cases}$$

Proposition 160 (Incidence factorisation of the Laplacian)

For any choice of edge orientations on an unweighted graph G, $L = B^T B$

Proof of Proposition 160 ($L = B^T B$) and worked check

Proof: The (u, v)-entry of $B^T B \in \mathcal{M}_n(\mathbb{R})$ is the dot product of columns u and v of B:

$$(B^T B)_{uv} = \sum_{k=1}^{m} b_{ku} b_{kv}$$

- If $u = v$: $b_{ku}^2 = (\pm 1)^2 = 1$ whenever edge e_k is incident to vertex u, and 0 otherwise, so $(B^T B)_{uu} = \sum_{k=1}^{m} b_{ku}^2 = d_u = \ell_{uu}$
- If $u \neq v$: the product $b_{ku} b_{kv}$ is non-zero only if edge e_k connects u and v, in which case one entry is $+1$ and the other is -1, so $b_{ku} b_{kv} = -1 = \ell_{uv}$ □

Check on the 4-vertex paw graph (orienting $e_1 = (1, 2), e_2 = (1, 3), e_3 = (2, 3), e_4 = (3, 4)$):

$$B = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix} \implies B^T B = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix} = L$$

Elementary matrix properties of L

From $L = D - A$ and $L = B^T B$, several useful matrix identities follow immediately:

1. **Trace of L:** By the handshaking lemma, the sum of the eigenvalues of L equals twice the total edge weight:

$$\sum_{k=1}^{n} \lambda_k = \operatorname{tr}(L) = \sum_{i=1}^{n} d_i = 2 \sum_{\{i,j\} \in E} a_{ij} \qquad (= 2|E| \text{ for an unweighted graph})$$

2. **Sum of rank-one edge matrices:** Each edge $e = \{i, j\} \in E$ contributes a 2×2 sub-Laplacian on vertices i and j. Letting $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n$ denote the standard basis of $\mathbb{R}^n$,

$$L = \sum_{\{i,j\} \in E} a_{ij} (\boldsymbol{e}_i - \boldsymbol{e}_j)(\boldsymbol{e}_i - \boldsymbol{e}_j)^T$$

3. **Weak diagonal dominance:** Every row i satisfies $|\ell_{ii}| = d_i = \sum_{j \neq i} a_{ij} = \sum_{j \neq i} |\ell_{ij}|$

The combinatorial graph Laplacian

Definition and basic examples

The Laplacian quadratic form

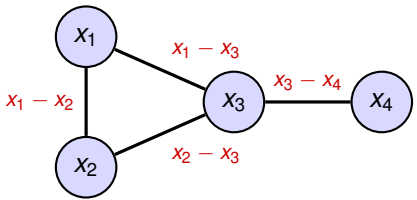
Vectors on $\mathbb{R}^n$ as functions on the vertices

How does L act on a vector $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)^T \in \mathbb{R}^n$?

Think of $\boldsymbol{x} \in \mathbb{R}^n$ as assigning a real number (a "signal" or "potential") $x_i \in \mathbb{R}$ to each vertex $i \in V$:

- At vertex i, the ith entry of $L\boldsymbol{x}$ is

$$(L\boldsymbol{x})_i = d_i x_i - \sum_{j=1}^{n} a_{ij} x_j = \sum_{j=1}^{n} a_{ij}(x_i - x_j)$$



- Along each edge $e_k = (i, j)$, the kth entry of $B\boldsymbol{x}$ is the **edge difference**

$$(B\boldsymbol{x})_k = x_i - x_j$$

Multiplying on the left by $\boldsymbol{x}^T$ produces the fundamental identity of spectral graph theory

The Laplacian quadratic form

Theorem 161 (Laplacian quadratic form)

Let $G = (V, E)$ be an undirected graph on n vertices with symmetric weight matrix $A = [a_{ij}]$ ($a_{ij} = a_{ji} \geq 0$) and combinatorial Laplacian $L = D - A$. Then for every vector $\boldsymbol{x} = (x_1, \ldots, x_n)^T \in \mathbb{R}^n$,

$$\boldsymbol{x}^T L \boldsymbol{x} = \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}(x_i - x_j)^2 = \sum_{\{i,j\} \in E} a_{ij}(x_i - x_j)^2$$

In particular, for an unweighted simple graph ($a_{ij} \in \{0, 1\}$),

$$\boldsymbol{x}^T L \boldsymbol{x} = \sum_{\{i,j\} \in E} (x_i - x_j)^2$$

Proof of Theorem 161 (Method 1: direct algebraic expansion)

Let $\boldsymbol{x} = (x_1, \ldots, x_n)^T \in \mathbb{R}^n$. Using $L = D - A$ and $d_i = \sum_{j=1}^n a_{ij}$, we expand

$$\boldsymbol{x}^T L \boldsymbol{x} = \boldsymbol{x}^T D \boldsymbol{x} - \boldsymbol{x}^T A \boldsymbol{x} = \sum_{i=1}^n d_i x_i^2 - \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j$$

$$= \sum_{i=1}^n \left(\sum_{j=1}^n a_{ij} \right) x_i^2 - \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j$$

By symmetry $a_{ij} = a_{ji}$, relabelling dummy indices $i \leftrightarrow j$ gives $\sum_{i,j} a_{ij} x_i^2 = \sum_{i,j} a_{ji} x_j^2 = \sum_{i,j} a_{ij} x_j^2$, so we can split the first sum into two equal halves:

$$\boldsymbol{x}^T L \boldsymbol{x} = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i^2 - \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j + \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_j^2$$

$$= \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n a_{ij} \left(x_i^2 - 2 x_i x_j + x_j^2 \right) = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n a_{ij} (x_i - x_j)^2$$

Since each undirected edge $\{i, j\} \in E$ appears twice in the double sum (as (i, j)

Proof of Theorem 161 (Method 2: via the incidence matrix)

For an unweighted graph $G = (V, E)$ with $m = |E|$ edges, Proposition 160 gives $L = B^T B$, where $B \in \mathcal{M}_{mn}$ is the oriented incidence matrix

For any $\boldsymbol{x} \in \mathbb{R}^n$, the kth entry of $B\boldsymbol{x} \in \mathbb{R}^m$ corresponding to oriented edge $e_k = (i, j)$ is

$$(B\boldsymbol{x})_k = x_i - x_j$$

Therefore,

$$\boldsymbol{x}^T L\boldsymbol{x} = \boldsymbol{x}^T(B^T B)\boldsymbol{x} = (B\boldsymbol{x})^T(B\boldsymbol{x}) = \|B\boldsymbol{x}\|_2^2 = \sum_{k=1}^{m}(B\boldsymbol{x})_k^2 = \sum_{\{i,j\}\in E}(x_i - x_j)^2$$

(For a weighted graph with diagonal edge-weight matrix $W_E = \mathrm{diag}(w_{e_1}, \ldots, w_{e_m})$, writing $L = B^T W_E B = (W_E^{1/2} B)^T (W_E^{1/2} B)$ gives $\boldsymbol{x}^T L\boldsymbol{x} = \|W_E^{1/2} B\boldsymbol{x}\|_2^2 = \sum_{\{i,j\}\in E} w_{ij}(x_i - x_j)^2$)

Geometric interpretation: smoothness (Dirichlet energy) on a graph

The quadratic form $\boldsymbol{x}^T L \boldsymbol{x} = \sum_{\{i,j\} \in E} a_{ij}(x_i - x_j)^2$ is called the **Dirichlet energy** of the vector $\boldsymbol{x}$ on the graph G:

- Every edge $\{i, j\} \in E$ acts like a spring with stiffness a_{ij} that penalises the squared difference $(x_i - x_j)^2$ between its endpoints
- **Smooth vector** ($\boldsymbol{x}^T L \boldsymbol{x}$ small): adjacent vertices have nearly equal values ($x_i \approx x_j$ whenever a_{ij} is large)
- **Constant vector** ($\boldsymbol{x}^T L \boldsymbol{x} = 0$): $x_i = x_j$ across every edge of G
- **Rough / oscillating vector** ($\boldsymbol{x}^T L \boldsymbol{x}$ large): adjacent vertices have very different values (e.g. alternating signs $+1$ and -1 across edges)

Example: evaluating $\boldsymbol{x}^T L\boldsymbol{x}$ on the 4-vertex paw graph

Take the 4-vertex paw graph with edges $\{1,2\},\{1,3\},\{2,3\},\{3,4\}$ and compare two vectors of equal norm:

Case 1 (smooth): $\boldsymbol{x} = (1,1,0,-1)^T$

$$\begin{aligned}\boldsymbol{x}^T L\boldsymbol{x} &= (x_1-x_2)^2 + (x_1-x_3)^2 \\ &\quad + (x_2-x_3)^2 + (x_3-x_4)^2 \\ &= (1-1)^2 + (1-0)^2 \\ &\quad + (1-0)^2 + (0-(-1))^2 \\ &= 0+1+1+1 = 3\end{aligned}$$

Adjacent vertices $\{1,2\}$ share the same value 1, and values change gradually along the graph

Case 2 (oscillating): $\boldsymbol{y} = (1,-1,0,1)^T$

$$\begin{aligned}\boldsymbol{y}^T L\boldsymbol{y} &= (y_1-y_2)^2 + (y_1-y_3)^2 \\ &\quad + (y_2-y_3)^2 + (y_3-y_4)^2 \\ &= (1-(-1))^2 + (1-0)^2 \\ &\quad + (-1-0)^2 + (0-1)^2 \\ &= 4+1+1+1 = 7\end{aligned}$$

Adjacent vertices 1 and 2 jump from $+1$ to -1, producing a much larger energy

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Fundamental spectral properties of the Laplacian

Theorem 162 (Fundamental spectral properties of L)

Let $G = (V, E)$ be an undirected graph on n vertices with symmetric nonnegative weight matrix $A = [a_{ij}]$ and combinatorial Laplacian $L = D - A$. Then:

1. *L is real and symmetric ($L^T = L$), so it has n real eigenvalues and an orthonormal basis of real eigenvectors $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$*
2. *L is **positive semi-definite** ($\boldsymbol{x}^T L \boldsymbol{x} \geq 0$ for all $\boldsymbol{x} \in \mathbb{R}^n$), so all eigenvalues of L are nonnegative and can be ordered in ascending order:*

$$0 \leq \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$$

3. *The all-ones vector $\mathbb{1} = (1, 1, \ldots, 1)^T$ satisfies $L\mathbb{1} = \mathbf{0}$, so $\lambda_1 = 0$ is always the smallest eigenvalue of L with eigenvector $\boldsymbol{v}_1 = \mathbb{1}$ (or unit eigenvector $\frac{1}{\sqrt{n}}\mathbb{1}$)*

Proof of Theorem 162

1. **Symmetry and real spectrum:** Since D is diagonal ($D^T = D$) and A is symmetric ($A^T = A$),

$$L^T = (D - A)^T = D^T - A^T = D - A = L$$

   By the spectral theorem for real symmetric matrices, L has n real eigenvalues and an orthonormal basis of real eigenvectors in $\mathbb{R}^n$

2. **Positive semi-definiteness and $\lambda_k \geq 0$:** By Theorem 161, for any $\boldsymbol{x} \in \mathbb{R}^n$,

$$\boldsymbol{x}^T L \boldsymbol{x} = \sum_{\{i,j\} \in E} a_{ij}(x_i - x_j)^2 \geq 0$$

   since $a_{ij} \geq 0$ and $(x_i - x_j)^2 \geq 0$. If $L\boldsymbol{v}_k = \lambda_k \boldsymbol{v}_k$ with $\boldsymbol{v}_k \neq \mathbf{0}$, then

$$\lambda_k = \frac{\boldsymbol{v}_k^T (L\boldsymbol{v}_k)}{\boldsymbol{v}_k^T \boldsymbol{v}_k} = \frac{\boldsymbol{v}_k^T L \boldsymbol{v}_k}{\|\boldsymbol{v}_k\|_2^2} \geq 0$$

3. **Zero row sums ($L\mathbb{1} = \mathbf{0}$):** For each row $i = 1, \ldots, n$,

$$(L\mathbb{1})_i = \sum_{j=1}^{n} \ell_{ij} \cdot 1 = d_i - \sum_{j=1}^{n} a_{ij} = d_i - d_i = 0 \implies L\mathbb{1} = \mathbf{0} = 0 \cdot \mathbb{1}$$ □

Beware of the eigenvalue ordering convention!

In PCA (Unit 05) and Perron-Frobenius theory (Units 08–09), we ordered eigenvalues from *largest to smallest* because we wanted maximum variance or the dominant long-term state

For the graph Laplacian L, the universal convention is the **ascending order**

$$0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$$

Why? Because for any unit eigenvector $\boldsymbol{v}_k$ ($\|\boldsymbol{v}_k\|_2 = 1$),

$$\lambda_k = \boldsymbol{v}_k^T L \boldsymbol{v}_k = \sum_{\{i,j\} \in E} a_{ij}(v_{ki} - v_{kj})^2$$

- **Smallest eigenvalues ($\lambda_1 = 0, \lambda_2, \ldots$)**: eigenvectors vary *least* across edges, revealing connected components and large-scale community structure
- **Largest eigenvalues ($\ldots, \lambda_n$)**: eigenvectors oscillate rapidly across edges

Multiplicity of $\lambda = 0$ and connected components

We know $\lambda_1 = 0$ is always an eigenvalue of L. When is $\lambda_2 = 0$ as well?

Theorem 163 (Multiplicity of 0 equals the number of connected components)

Let $G = (V, E)$ be an undirected graph with nonnegative edge weights and Laplacian L. The multiplicity of 0 *as an eigenvalue of L (that is, $\text{nullity}(L) = \dim(\text{null}(L))$) equals the number k of connected components $C_1, C_2, \ldots, C_k$ of G*

Moreover, the component indicator vectors $\mathbb{1}_{C_1}, \mathbb{1}_{C_2}, \ldots, \mathbb{1}_{C_k} \in \{0,1\}^n$, defined by

$$(\mathbb{1}_{C_r})_i = \begin{cases} 1 & \text{if } i \in C_r \\ 0 & \text{otherwise} \end{cases}$$

form an orthogonal basis of $\text{null}(L)$

Proof of Theorem 163 – Part 1: connected graph ($k = 1$)

Suppose first that G is connected ($k = 1$). We must show that $\text{null}(L) = \text{span}\{\mathbb{1}\}$, so $\dim(\text{null}(L)) = 1$

1. We already know $L\mathbb{1} = \mathbf{0}$, so $\text{span}\{\mathbb{1}\} \subseteq \text{null}(L)$
2. Conversely, let $\boldsymbol{x} \in \text{null}(L)$, so $L\boldsymbol{x} = \mathbf{0}$. Taking the dot product with $\boldsymbol{x}$ and applying Theorem 161 gives
$$0 = \boldsymbol{x}^T\mathbf{0} = \boldsymbol{x}^T L\boldsymbol{x} = \sum_{\{i,j\}\in E} a_{ij}(x_i - x_j)^2$$
3. For every edge $\{i, j\} \in E$, we have $a_{ij} > 0$ and $(x_i - x_j)^2 \geq 0$. A sum of nonnegative terms equals 0 if and only if every term is 0:
$$(x_i - x_j)^2 = 0 \implies x_i = x_j \quad \text{for every edge } \{i, j\} \in E$$
4. Since G is connected, any two vertices $u, v \in V$ are linked by a path $u = i_0, i_1, \ldots, i_m = v$ in G, so $x_u = x_{i_1} = \cdots = x_v$
5. Hence all entries of $\boldsymbol{x}$ are equal ($x_1 = \cdots = x_n = c$), i.e. $\boldsymbol{x} = c\,\mathbb{1} \in \text{span}\{\mathbb{1}\}$

□

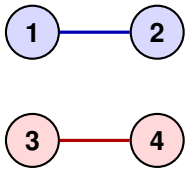
Proof of Theorem 163 – Part 2: k connected components ($k \geq 2$)

Now suppose G has k connected components $C_1, \ldots, C_k$. Reorder the vertices so that those in C_1 come first, followed by $C_2, \ldots, C_k$

1. Because there are no edges between different components, A, D and L are block-diagonal:
$$L = \begin{pmatrix} L_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & L_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & L_k \end{pmatrix}$$
where each L_r is the Laplacian of the connected subgraph on C_r
2. Writing $\boldsymbol{x} = (\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(k)})^T$ according to the blocks, $L\boldsymbol{x} = \mathbf{0} \iff L_r \boldsymbol{x}^{(r)} = \mathbf{0}$ for every $r = 1, \ldots, k$
3. By Part 1 applied to each connected component C_r, $L_r \boldsymbol{x}^{(r)} = \mathbf{0} \iff \boldsymbol{x}^{(r)} = c_r \mathbb{1}_{|C_r|}$, which in $\mathbb{R}^n$ means
$$\boldsymbol{x} = c_1 \mathbb{1}_{C_1} + c_2 \mathbb{1}_{C_2} + \cdots + c_k \mathbb{1}_{C_k}$$
4. Since $C_1, \ldots, C_k$ are disjoint, $\mathbb{1}_{C_r}^T \mathbb{1}_{C_s} = 0$ for $r \neq s$, so $\{\mathbb{1}_{C_1}, \ldots, \mathbb{1}_{C_k}\}$ is an orthogonal basis of $\text{null}(L)$ and $\dim(\text{null}(L)) = k$

Example: a disconnected graph on 4 vertices

Consider $G = (V, E)$ on $n = 4$ vertices with only two edges $\{1, 2\}$ and $\{3, 4\}$, so G has $k = 2$ connected components $C_1 = \{1, 2\}$ and $C_2 = \{3, 4\}$:



$$L = \left(\begin{array}{cc|cc} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ \hline 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{array}\right) = \begin{pmatrix} L_1 & \mathbf{0} \\ \mathbf{0} & L_2 \end{pmatrix}$$

- Each 2×2 block $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ has trace 2 and determinant 0, so its eigenvalues are 0 and 2
- The spectrum of L is therefore $\lambda_1 = 0,\ \lambda_2 = 0,\ \lambda_3 = 2,\ \lambda_4 = 2$ (multiplicity of 0 is $k = 2$)
- The orthogonal indicator vectors for null(L) are

$$\mathbb{1}_{C_1} = (1, 1, 0, 0)^T \qquad \text{and} \qquad \mathbb{1}_{C_2} = (0, 0, 1, 1)^T$$

Counting connected components with $\lambda = 0$ in R

Let us build a graph on $n = 8$ vertices consisting of $k = 3$ disconnected components (a triangle $\{1, 2, 3\}$, a path $\{4, 5, 6\}$ and an edge $\{7, 8\}$) and check the eigenvalues of L:

```
G_comp = make_graph(edges = c(1,2, 2,3, 1,3,  4,5, 5,6,  7,8),
                    n = 8, directed = FALSE)
L_comp = as.matrix(laplacian_matrix(G_comp))
ev_comp = eigen(L_comp, symmetric = TRUE)
# Eigenvalues of L in ascending order
round(rev(ev_comp$values), 4)

## [1] 0 0 0 1 2 3 3 3

# Number of zero eigenvalues (up to numerical precision)
sum(abs(ev_comp$values) < 1e-10)

## [1] 3
```

Exactly three eigenvalues of L equal 0, matching the $k = 3$ connected

Spectrum of extreme graphs: empty vs complete graph K_n

Comparing two extreme graphs on n vertices illustrates how Laplacian eigenvalues measure connectivity:

1. Empty graph E_n ($|E| = 0$):

$$L = \mathbf{0}_n$$

- n isolated vertices ($k = n$ components)
- Spectrum of L:

$$\lambda_1 = \lambda_2 = \cdots = \lambda_n = 0$$

2. Complete graph K_n (all pairs connected):

$$L = (n-1)\mathbb{I}_n - (\mathbb{J}_n - \mathbb{I}_n) = n\mathbb{I}_n - \mathbb{1}\mathbb{1}^T$$

- For $\boldsymbol{v}_1 = \mathbb{1}$:
  $L\mathbb{1} = n\mathbb{1} - n\mathbb{1} = \mathbf{0} \implies \lambda_1 = 0$
- For any $\boldsymbol{x} \perp \mathbb{1}$ ($\mathbb{1}^T\boldsymbol{x} = 0$):

$$L\boldsymbol{x} = n\boldsymbol{x} - \mathbb{1}(\mathbb{1}^T\boldsymbol{x}) = n\boldsymbol{x}$$

- Spectrum of K_n: $\lambda_1 = 0$ and $\lambda_2 = \cdots = \lambda_n = n$

For every simple graph G on n vertices, its eigenvalues lie between these two extremes: $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n \leq n$

What if G is "almost" disconnected?

Suppose G has two dense clusters C_1 and C_2 that are completely disconnected ($k = 2$):

- Then $\lambda_1 = 0$ and $\lambda_2 = 0$, and any vector in null(L) is constant on C_1 and constant on C_2:

$$\boldsymbol{v}_1 = (1, \ldots, 1,\ 1, \ldots, 1)^T, \qquad \boldsymbol{v}_2 = (1, \ldots, 1,\ -1, \ldots, -1)^T \quad (\text{if } |C_1| = |C_2|)$$

Now perturb G slightly by adding a single weak bridge edge between C_1 and C_2:

- The graph is now connected ($k = 1$), so $\lambda_1 = 0$ and $\lambda_2 > 0$
- Because eigenvalues and eigenvectors of symmetric matrices vary continuously under small perturbations, λ_2 will be **positive but very small** ($\lambda_2 \approx 0$)
- The associated eigenvector $\boldsymbol{v}_2 \perp \mathbb{1}$ will remain **nearly constant and positive on** C_1 and **nearly constant and negative on** C_2!

Next, we make this intuition completely rigorous via **graph cuts**

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Graph cuts, the Fiedler vector and spectral clustering

- **Graph partitioning and cut size**
- Spectral relaxation and the Fiedler vector
- Worked exam-style examples and R implementation

Partitions and the cut of a graph

Let $G = (V, E)$ be an undirected graph on n vertices with symmetric edge weights $a_{ij} \geq 0$

A **2-way partition** (or **bipartition**) of G splits V into two disjoint non-empty subsets S and $\bar{S} = V \setminus S$ ($S \cup \bar{S} = V$, $S \cap \bar{S} = \emptyset$)

Definition 164 (Cut of a partition)

*The **cut** between S and $\bar{S}$ is the total weight of edges with one endpoint in S and the other endpoint in $\bar{S}$:*

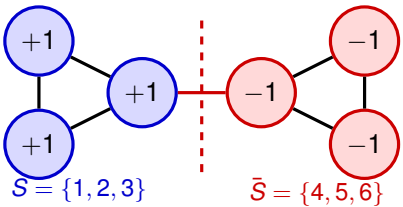
$$\text{cut}(S, \bar{S}) = \sum_{i \in S,\, j \in \bar{S}} a_{ij}$$

For an unweighted graph ($a_{ij} \in \{0, 1\}$), $\text{cut}(S, \bar{S})$ is simply the number of edges crossing between S and $\bar{S}$

Encoding a partition with a ± 1 indicator vector

To connect graph cuts to linear algebra, we represent any bipartition $(S, \bar{S})$ by a **cut indicator vector** $\boldsymbol{x} = (x_1, \ldots, x_n)^T \in \{-1, +1\}^n$:

$$x_i = \begin{cases} +1 & \text{if } i \in S \\ -1 & \text{if } i \in \bar{S} \end{cases}$$



Proposition 165 (Cut size as a quadratic form)

For any partition $(S, \bar{S})$ of V with indicator vector $\boldsymbol{x} \in \{-1, +1\}^n$,

$$\boldsymbol{x}^T L \boldsymbol{x} = 4 \operatorname{cut}(S, \bar{S})$$

or equivalently,

$$\operatorname{cut}(S, \bar{S}) = \frac{1}{4} \boldsymbol{x}^T L \boldsymbol{x}$$

Proof of Proposition 165 ($\boldsymbol{x}^T L \boldsymbol{x} = 4\,\mathrm{cut}(S, \bar{S})$)

Let $\boldsymbol{x} \in \{-1, +1\}^n$ be the cut indicator vector for $(S, \bar{S})$. For any edge $\{i, j\} \in E$, look at the squared difference $(x_i - x_j)^2$:

- **Both endpoints in** S ($i \in S, j \in S$): $x_i = +1$ and $x_j = +1$, so
$$(x_i - x_j)^2 = (1 - 1)^2 = 0$$
- **Both endpoints in** $\bar{S}$ ($i \in \bar{S}, j \in \bar{S}$): $x_i = -1$ and $x_j = -1$, so
$$(x_i - x_j)^2 = (-1 - (-1))^2 = 0$$
- **Endpoints in different sets** ($i \in S, j \in \bar{S}$ or $i \in \bar{S}, j \in S$): $x_i - x_j = \pm 2$, so
$$(x_i - x_j)^2 = (\pm 2)^2 = 4$$

Substituting into Theorem 161 ($\boldsymbol{x}^T L \boldsymbol{x} = \sum_{\{i,j\} \in E} a_{ij}(x_i - x_j)^2$), only the edges crossing between S and $\bar{S}$ survive, each multiplied by 4:

$$\boldsymbol{x}^T L \boldsymbol{x} = \sum_{i \in S,\, j \in \bar{S}} 4\, a_{ij} = 4 \sum_{i \in S,\, j \in \bar{S}} a_{ij} = 4\,\mathrm{cut}(S, \bar{S}) \qquad \square$$

Example: comparing two cuts on the 4-vertex paw graph

Recall the 4-vertex paw graph with Laplacian

$$L = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

Let us compare two candidate partitions using $\text{cut}(S, \bar{S}) = \frac{1}{4}\boldsymbol{x}^T L \boldsymbol{x}$:

Partition A: $S_1 = \{1, 2, 3\}$, $\bar{S}_1 = \{4\}$

$$\boldsymbol{x}^{(1)} = (1, 1, 1, -1)^T$$

$$L\boldsymbol{x}^{(1)} = \begin{pmatrix} 0 \\ 0 \\ 2 \\ -2 \end{pmatrix} \implies (\boldsymbol{x}^{(1)})^T L \boldsymbol{x}^{(1)} = 4$$

$$\text{cut}(S_1, \bar{S}_1) = \frac{4}{4} = 1 \quad \text{(only edge } \{3, 4\}\text{)}$$

Partition B: $S_2 = \{1, 3\}$, $\bar{S}_2 = \{2, 4\}$

$$\boldsymbol{x}^{(2)} = (1, -1, 1, -1)^T$$

$$L\boldsymbol{x}^{(2)} = \begin{pmatrix} 2 \\ -4 \\ 4 \\ -2 \end{pmatrix} \implies (\boldsymbol{x}^{(2)})^T L \boldsymbol{x}^{(2)} = 12$$

$$\text{cut}(S_2, \bar{S}_2) = \frac{12}{4} = 3 \quad \text{(edges } \{1, 2\}, \{2, 3\},$$

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Why unconstrained minimum cut is not enough

Since $\text{cut}(S, \bar{S}) = \frac{1}{4}\boldsymbol{x}^T L \boldsymbol{x}$, can we cluster a graph simply by minimising $\boldsymbol{x}^T L \boldsymbol{x}$ over $\boldsymbol{x} \in \{-1, +1\}^n$?

1. **Trivial all-in-one solution:** Without constraints, the minimum of $\boldsymbol{x}^T L \boldsymbol{x}$ over $\{-1, +1\}^n$ is 0, achieved at $\boldsymbol{x} = +\mathbb{1}$ ($S = V, \bar{S} = \emptyset$) and $\boldsymbol{x} = -\mathbb{1}$ ($S = \emptyset, \bar{S} = V$)

2. **Unbalanced "whisker" cuts:** Even if we forbid $\boldsymbol{x} = \pm\mathbb{1}$, minimum cut often slices off a single degree-1 vertex (like vertex 4 in the paw graph) rather than splitting the graph into two meaningful communities of comparable size

To find balanced communities, we must add a **balance constraint** requiring $|S|$ and $|\bar{S}|$ to be equal (or comparable) in size

Encoding balanced bisection algebraically

Suppose n is even and we want a **balanced bisection** with $|S| = |\bar{S}| = n/2$

For any cut indicator vector $\boldsymbol{x} \in \{-1, +1\}^n$, notice two algebraic facts:

1. **Balance equals orthogonality to $\mathbb{1}$:** Since $x_i = +1$ on S and $x_i = -1$ on $\bar{S}$,

$$\mathbb{1}^T \boldsymbol{x} = \sum_{i=1}^{n} x_i = |S| - |\bar{S}|$$

   Therefore,

$$|S| = |\bar{S}| \quad \iff \quad \mathbb{1}^T \boldsymbol{x} = 0 \quad \iff \quad \boldsymbol{x} \perp \mathbb{1}$$

2. **Fixed Euclidean norm:** Every vector $\boldsymbol{x} \in \{-1, +1\}^n$ has squared norm

$$\|\boldsymbol{x}\|_2^2 = \boldsymbol{x}^T \boldsymbol{x} = \sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} (\pm 1)^2 = n$$

General balanced cuts: RatioCut for unequal cluster sizes

What if the two true clusters do not have the exact same size ($|S| \neq |\bar{S}|$)? We penalise small sets by minimising the **RatioCut** (Hagen & Kahng, 1992):

$$\text{RatioCut}(S, \bar{S}) = \text{cut}(S, \bar{S}) \left(\frac{1}{|S|} + \frac{1}{|\bar{S}|} \right) = \frac{n}{|S|\,|\bar{S}|} \text{cut}(S, \bar{S})$$

Define the weighted indicator vector $\boldsymbol{x} = (x_1, \ldots, x_n)^T$ by

$$x_i = \begin{cases} +\sqrt{|\bar{S}|/|S|} & \text{if } i \in S \\ -\sqrt{|S|/|\bar{S}|} & \text{if } i \in \bar{S} \end{cases}$$

Then one checks directly that:

1. $\mathbb{1}^T \boldsymbol{x} = |S|\sqrt{|\bar{S}|/|S|} - |\bar{S}|\sqrt{|S|/|\bar{S}|} = \sqrt{|S|\,|\bar{S}|} - \sqrt{|S|\,|\bar{S}|} = 0$ ($\boldsymbol{x} \perp \mathbb{1}$)
2. $\|\boldsymbol{x}\|_2^2 = |S|\frac{|\bar{S}|}{|S|} + |\bar{S}|\frac{|S|}{|\bar{S}|} = |\bar{S}| + |S| = n$, and $\boldsymbol{x}^T L \boldsymbol{x} = n\,\text{RatioCut}(S, \bar{S})$

From NP-hard discrete cut to continuous spectral relaxation

Both balanced minimum bisection and RatioCut lead to minimising $\boldsymbol{x}^T L \boldsymbol{x}$ subject to $\boldsymbol{x} \perp \mathbb{1}$, $\|\boldsymbol{x}\|_2^2 = n$ and a discrete two-value constraint on the entries x_i:

$$\min_{\substack{\boldsymbol{x} \in \{-1,+1\}^n \\ \boldsymbol{x}^T \mathbb{1} = 0}} \frac{1}{4} \boldsymbol{x}^T L \boldsymbol{x}$$

- **Computational barrier:** Checking all balanced partitions $\binom{n}{n/2}$ is **NP-hard** (for $n = 100$ vertices, $\binom{100}{50} \approx 1.0 \times 10^{29}$)
- **Spectral relaxation:** Relax the discrete constraint on x_i to allow *arbitrary real values* $\boldsymbol{x} \in \mathbb{R}^n$ that still satisfy the norm constraint $\boldsymbol{x}^T \boldsymbol{x} = n$ and the balance constraint $\boldsymbol{x}^T \mathbb{1} = 0$:

$$\min_{\substack{\boldsymbol{x} \in \mathbb{R}^n \\ \boldsymbol{x}^T \boldsymbol{x} = n,\ \boldsymbol{x}^T \mathbb{1} = 0}} \boldsymbol{x}^T L \boldsymbol{x} \quad \Longleftrightarrow \quad \min_{\substack{\boldsymbol{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\} \\ \boldsymbol{x} \perp \mathbb{1}}} \frac{\boldsymbol{x}^T L \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}}$$

Solving the relaxed problem via the Rayleigh quotient

Theorem 166 (Rayleigh quotient theorem for λ_2)

Let L be the Laplacian of an undirected graph G on n vertices with eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ and orthonormal eigenvectors $\boldsymbol{v}_1 = \frac{1}{\sqrt{n}}\mathbb{1}, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n$. Then

$$\min_{\substack{\boldsymbol{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\} \\ \boldsymbol{x} \perp \mathbb{1}}} \frac{\boldsymbol{x}^T L \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}} = \lambda_2$$

and the minimum is attained when $\boldsymbol{x}$ is any non-zero scalar multiple of $\boldsymbol{v}_2$

In particular, if we scale $\boldsymbol{v}_2$ so that $\|\boldsymbol{x}\|_2^2 = n$, the optimal continuous vector is $\boldsymbol{x} = \sqrt{n}\,\boldsymbol{v}_2$, with objective value $\boldsymbol{x}^T L \boldsymbol{x} = n\lambda_2$ (giving the lower bound $\text{cut}(S, \bar{S}) \geq \frac{n}{4}\lambda_2$ for any balanced bisection!)

Proof of Theorem 166

Let $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ be an orthonormal basis of eigenvectors of L with $L\boldsymbol{v}_k = \lambda_k \boldsymbol{v}_k$, where $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ and $\boldsymbol{v}_1 = \frac{1}{\sqrt{n}}\mathbb{1}$

1. Expand any vector $\boldsymbol{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ in the orthonormal basis $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$:

$$\boldsymbol{x} = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + \cdots + c_n\boldsymbol{v}_n, \qquad c_k = \boldsymbol{v}_k^T\boldsymbol{x}$$

2. The orthogonality constraint $\boldsymbol{x} \perp \mathbb{1}$ means $c_1 = \boldsymbol{v}_1^T\boldsymbol{x} = \frac{1}{\sqrt{n}}\mathbb{1}^T\boldsymbol{x} = 0$, so

$$\boldsymbol{x} = \sum_{k=2}^{n} c_k\boldsymbol{v}_k \qquad \text{and} \qquad \boldsymbol{x}^T\boldsymbol{x} = \sum_{k=2}^{n} c_k^2 > 0$$

3. Since $L\boldsymbol{x} = \sum_{k=2}^{n} c_k\lambda_k\boldsymbol{v}_k$ and the $\boldsymbol{v}_k$ are orthonormal,

$$\frac{\boldsymbol{x}^T L\boldsymbol{x}}{\boldsymbol{x}^T\boldsymbol{x}} = \frac{\sum_{k=2}^{n}\lambda_k c_k^2}{\sum_{k=2}^{n} c_k^2} \geq \frac{\sum_{k=2}^{n}\lambda_2 c_k^2}{\sum_{k=2}^{n} c_k^2} = \lambda_2$$

with equality when $c_2 \neq 0$ and $c_3 = \cdots = c_n = 0$, i.e. $\boldsymbol{x} = c_2\boldsymbol{v}_2$ □

Algebraic connectivity and the Fiedler vector

Miroslav Fiedler (1973) discovered the fundamental role of λ_2 and $\boldsymbol{v}_2$ in measuring how well-connected a graph is

Definition 167 (Algebraic connectivity and Fiedler vector)

Let $G = (V, E)$ be an undirected graph on n vertices with Laplacian eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$

- *The second smallest eigenvalue λ_2 is called the **algebraic connectivity** (or **Fiedler value**) of G*
- *An eigenvector $\boldsymbol{v}_2$ associated with λ_2 (orthogonal to $\mathbb{1}$) is called a **Fiedler vector** of G*

- By Theorem 163, $\lambda_2 > 0$ **if and only if** G **is connected**
- Because $\boldsymbol{v}_2 \perp \mathbb{1}$, the entries of $\boldsymbol{v}_2$ sum to 0 ($\sum_{i=1}^{n} v_{2i} = 0$), so $\boldsymbol{v}_2$ always has **both positive and negative entries**!

The 2-way spectral partitioning algorithm

How do we turn the continuous Fiedler vector $\boldsymbol{v}_2 = (v_{21}, \ldots, v_{2n})^T \in \mathbb{R}^n$ back into a discrete partition $(S, \bar{S})$? We round its entries by their signs!

Algorithm: 2-way unnormalised spectral clustering (Fiedler bisection)

Input: Undirected graph $G = (V, E)$ with symmetric adjacency/similarity matrix $A \in \mathcal{M}_n(\mathbb{R}_{\geq 0})$

1. Compute the degree matrix $D = \operatorname{diag}(A\mathbb{1})$ and the Laplacian $L = D - A$
2. Compute the second smallest eigenvalue λ_2 and associated Fiedler vector $\boldsymbol{v}_2 = (v_{21}, \ldots, v_{2n})^T$ of L
3. Partition the vertices according to the sign of v_{2i}:

$$S = \{i \in V : v_{2i} \geq 0\}, \qquad \bar{S} = \{i \in V : v_{2i} < 0\}$$

(Alternatively, split at the median of $\boldsymbol{v}_2$ if an exact $n/2 - n/2$ split is required)

Note: Because $-\boldsymbol{v}_2$ is also a Fiedler vector, flipping the sign of $\boldsymbol{v}_2$ simply swaps the labels $S \leftrightarrow \bar{S}$, which defines the *exact same* partition of V

Extension to k clusters and normalised Laplacians

What if a graph (or dataset) has $k > 2$ clusters, or vertices with wildly different degrees?

- **k-way spectral clustering:**
  1. Compute the first k eigenvectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_k$ of L associated with the k smallest eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_k$
  2. Stack them as columns of the matrix $U = [\boldsymbol{v}_1 \;\; \boldsymbol{v}_2 \;\; \cdots \;\; \boldsymbol{v}_k] \in \mathcal{M}_{nk}(\mathbb{R})$
  3. Treat row i of U as the new embedding $\boldsymbol{y}_i \in \mathbb{R}^k$ of vertex i, and run k-means on $\{\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n\} \subset \mathbb{R}^k$

- **Normalised graph Laplacians** (used when degrees d_i vary over orders of magnitude):

$$L_{\text{sym}} = D^{-1/2} L D^{-1/2} = \mathbb{I} - D^{-1/2} A D^{-1/2}, \qquad L_{\text{rw}} = D^{-1} L = \mathbb{I} - D^{-1} A$$

  Notice that $D^{-1}A$ is the random-walk transition matrix on G! Spectral clustering with L_{sym} or L_{rw} relaxes the degree-normalised cut (Shi & Malik 2000; Ng, Jordan & Weiss 2001)

Choosing the number of clusters k: the eigengap heuristic

How do we choose the number of clusters k from the spectrum $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$?

- By Theorem 163, if G consists of k completely disconnected clusters, then

$$\lambda_1 = \lambda_2 = \cdots = \lambda_k = 0 \qquad \text{and} \qquad \lambda_{k+1} > 0$$

- When those k dense clusters are joined by only a few weak between-cluster edges, the first k eigenvalues remain **very close to** 0, while λ_{k+1} jumps sharply upward
- **Eigengap heuristic**: plot $\lambda_1, \lambda_2, \ldots, \lambda_m$ in ascending order and look for the index k with the largest **eigengap**

$$\Delta_k = \lambda_{k+1} - \lambda_k$$

For example, if $\lambda_1 = 0,\ \lambda_2 = 0.04,\ \lambda_3 = 0.07,\ \lambda_4 = 2.85,\ \lambda_5 = 3.10$, the large gap $\Delta_3 = \lambda_4 - \lambda_3 = 2.78$ strongly indicates $k = 3$ clusters

Graph cuts, the Fiedler vector and spectral clustering

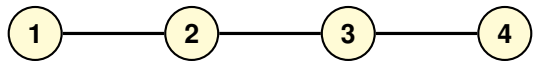
Graph partitioning and cut size

Spectral relaxation and the Fiedler vector

Worked exam-style examples and R implementation

Example 1: spectral bisection of the 4-vertex path P_4

Consider the 4-vertex path graph P_4:



Step 1 – Degree, adjacency and Laplacian matrices:

$$D = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \implies L = D - A = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

Example 1 (continued): Fiedler vector and cut of P_4

Step 2 – Verify the Fiedler eigenpair $(\lambda_2, \boldsymbol{v}_2)$: Claim: $\lambda_2 = 2 - \sqrt{2} \approx 0.586$ and $\boldsymbol{v}_2 = \begin{pmatrix} 1, & \sqrt{2}-1, & 1-\sqrt{2}, & -1 \end{pmatrix}^T \approx \begin{pmatrix} 1, & 0.414, & -0.414, & -1 \end{pmatrix}^T$
Let us verify $L\boldsymbol{v}_2 = \lambda_2 \boldsymbol{v}_2$ directly:

$$L\boldsymbol{v}_2 = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ \sqrt{2}-1 \\ 1-\sqrt{2} \\ -1 \end{pmatrix} = \begin{pmatrix} 2-\sqrt{2} \\ 3\sqrt{2}-4 \\ 4-3\sqrt{2} \\ \sqrt{2}-2 \end{pmatrix} = (2-\sqrt{2}) \begin{pmatrix} 1 \\ \sqrt{2}-1 \\ 1-\sqrt{2} \\ -1 \end{pmatrix} = \lambda_2 \boldsymbol{v}_2$$

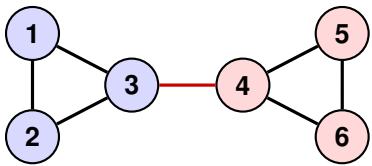
(since $(2-\sqrt{2})(\sqrt{2}-1) = 3\sqrt{2}-4$). Notice also that
$\mathbb{1}^T \boldsymbol{v}_2 = 1 + (\sqrt{2}-1) + (1-\sqrt{2}) - 1 = 0$

Step 3 – Partition by the signs of $\boldsymbol{v}_2$:

- $v_{21} = 1 > 0$ and $v_{22} = \sqrt{2} - 1 \approx 0.414 > 0 \implies S = \{1, 2\}$
- $v_{23} = 1 - \sqrt{2} \approx -0.414 < 0$ and $v_{24} = -1 < 0 \implies \bar{S} = \{3, 4\}$
- The spectral cut severing the middle edge $\{2, 3\}$ has $\text{cut}(S, \bar{S}) = 1$

Example 2: the 6-vertex barbell graph

Consider two triangles $C_1 = \{1, 2, 3\}$ and $C_2 = \{4, 5, 6\}$ joined by a single bridge edge $\{3, 4\}$:



Degrees: $d_1 = d_2 = d_5 = d_6 = 2$, $d_3 = d_4 = 3$

$$L = \left(\begin{array}{ccc|ccc} 2 & -1 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 \\ -1 & -1 & 3 & -1 & 0 & 0 \\ \hline 0 & 0 & -1 & 3 & -1 & -1 \\ 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & -1 & 2 \end{array}\right)$$

Notice that L is nearly block-diagonal: the two 3×3 blocks are coupled only through $\ell_{34} = \ell_{43} = -1$ and the extra $+1$ on ℓ_{33} and ℓ_{44}

Example 2 (continued): exact Fiedler vector of the barbell

By symmetry between $\{1,2\}$ and $\{5,6\}$ and antisymmetry across the bridge $\{3,4\}$ (so that $\mathbb{1}^T \boldsymbol{v}_2 = 0$), we look for a Fiedler vector of the form

$$\boldsymbol{v}_2 = (1,\ 1,\ c,\ -c,\ -1,\ -1)^T$$

Substitute into $L\boldsymbol{v}_2 = \lambda \boldsymbol{v}_2$:

- **Row 1 (and Row 2):** $2(1) - 1(1) - 1(c) = 1 - c = \lambda \implies c = 1 - \lambda$
- **Row 3:** $-1(1) - 1(1) + 3c - 1(-c) = 4c - 2 = \lambda c$

Substitute $c = 1 - \lambda$ into $4c - 2 = \lambda c$:

$$4(1-\lambda) - 2 = \lambda(1-\lambda) \iff \lambda^2 - 5\lambda + 2 = 0 \implies \lambda_2 = \frac{5 - \sqrt{17}}{2} \approx 0.4384$$

Then $c = 1 - \lambda_2 = \frac{\sqrt{17}-3}{2} \approx 0.5616 > 0$, giving the Fiedler vector

$$\boldsymbol{v}_2 \approx (1,\ 1,\ 0.562,\ -0.562,\ -1,\ -1)^T$$

Splitting by sign (≥ 0 vs < 0) gives $S = \{1,2,3\}$ and $\bar{S} = \{4,5,6\}$, cutting only the bridge $\{3,4\}$!

Computing the barbell Fiedler vector in R

We can build the barbell Laplacian in R (either from $D - A$ or with `igraph::laplacian_matrix`) and inspect its eigenvalues and Fiedler vector:

```
A_barbell = matrix(c(
  0, 1, 1, 0, 0, 0,
  1, 0, 1, 0, 0, 0,
  1, 1, 0, 1, 0, 0,
  0, 0, 1, 0, 1, 1,
  0, 0, 0, 1, 0, 1,
  0, 0, 0, 1, 1, 0
), nrow = 6, byrow = TRUE)
L_barbell = diag(rowSums(A_barbell)) - A_barbell
ev_b = eigen(L_barbell, symmetric = TRUE)
round(rev(ev_b$values), 4)          # eigenvalues in ascending order

## [1] 0.0000 0.4384 3.0000 3.0000 3.0000 4.5616

round(ev_b$vectors[, 5], 4)         # unit Fiedler vector v_2
```

Real-world benchmark: Zachary's Karate club network

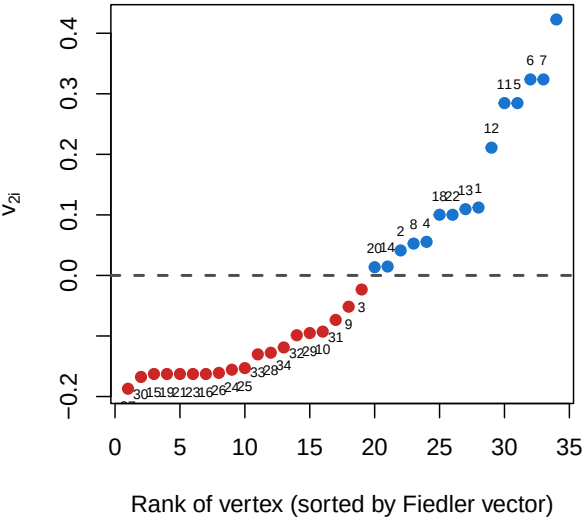
Zachary (1977) recorded social interactions among $n = 34$ members of a university karate club just before a dispute between the instructor (Mr Hi, vertex 1) and the club president (John A, vertex 34) split the club into two factions

```
G_karate = make_graph("Zachary")
L_karate = as.matrix(laplacian_matrix(G_karate))
ev_k = eigen(L_karate, symmetric = TRUE)
n_k = vcount(G_karate)
# Smallest three eigenvalues of L
round(ev_k$values[n_k:(n_k - 2)], 4)

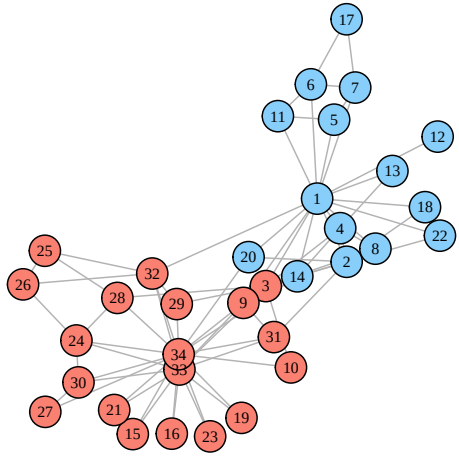
## [1] 0.0000 0.4685 0.9092

# Fiedler vector v_2 (oriented so vertex 1 is positive)
v2_k = ev_k$vectors[, n_k - 1]
if (v2_k[1] < 0) v2_k = -v2_k
table(Sign_of_v2 = ifelse(v2_k >= 0, "Group 1 (+)", "Group 2 (-)"))
```

Sorted entries of the Fiedler vector



Karate club partitioned by sign(v2)



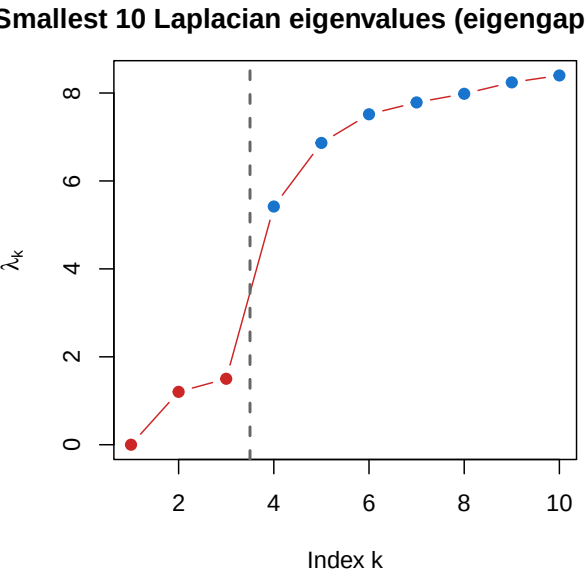
k-way spectral clustering in R ($k = 3$ communities)

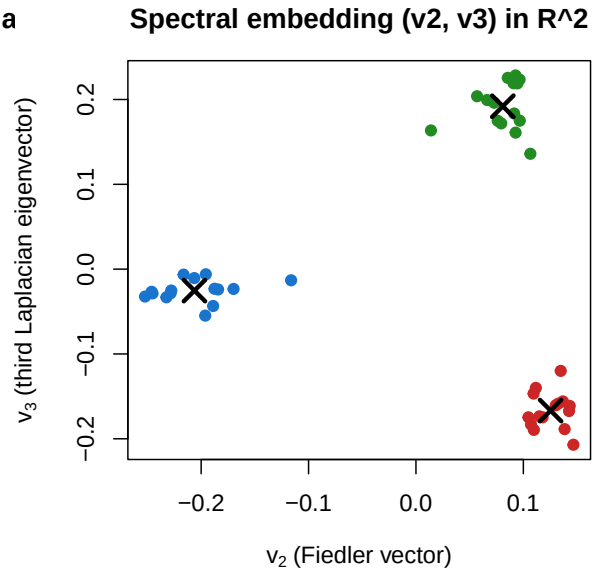
Suppose a network has $k = 3$ dense communities of 15 vertices each ($n = 45$), with high within-community edge probability ($p_{\text{in}} = 0.7$) and low between-community probability ($p_{\text{out}} = 0.03$):

```
set.seed(2740)
pm = matrix(0.03, nrow = 3, ncol = 3); diag(pm) = 0.70
G_sbm = sample_sbm(45, pref.matrix = pm, block.sizes = c(15, 15, 15))
L_sbm = as.matrix(laplacian_matrix(G_sbm))
ev_sbm = eigen(L_sbm, symmetric = TRUE)
# Smallest 6 eigenvalues of L (note the jump after lambda_3!)
round(rev(ev_sbm$values)[1:6], 3)

## [1] 0.000 1.203 1.500 5.418 6.865 7.517

# Spectral embedding using v_2 and v_3, followed by k-means (k = 3)
U_sbm = ev_sbm$vectors[, c(44, 43)]
km_sbm = kmeans(U_sbm, centers = 3, nstart = 20)
table(True_block = rep(1:3, each = 15), Spectral_cluster = km_sbm$cluster)
```


Smallest 10 Laplacian eigenvalues (eigengap a
λk
0
2
4
6
8
2
4
6
8
10
Index k


Spectral embedding (v2, v3) in R^2
v3 (third Laplacian eigenvector)
0.2
0.1
0.0
−0.1
−0.2
−0.2
−0.1
0.0
0.1
v2 (Fiedler vector)

Summary of Unit 10

- For an undirected graph $G = (V, E)$ with symmetric adjacency/weight matrix A and degree matrix $D = \mathrm{diag}(A\mathbb{1})$, the **combinatorial Laplacian** is $L = D - A = B^T B$
- The **Laplacian quadratic form** measures smoothness across edges:

$$\boldsymbol{x}^T L \boldsymbol{x} = \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}(x_i - x_j)^2 = \sum_{\{i,j\} \in E} a_{ij}(x_i - x_j)^2 \geq 0$$

- L is real, symmetric and positive semi-definite with eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ and $L\mathbb{1} = \mathbf{0}$; the multiplicity of $\lambda = 0$ equals the number of connected components of G
- For any ± 1 cut indicator vector $\boldsymbol{x} \in \{-1, +1\}^n$, $\mathrm{cut}(S, \bar{S}) = \frac{1}{4} \boldsymbol{x}^T L \boldsymbol{x}$
- Relaxing balanced minimum cut ($\boldsymbol{x} \perp \mathbb{1}$, $\|\boldsymbol{x}\|_2^2 = n$) to $\mathbb{R}^n$ yields the **Fiedler value** λ_2 and **Fiedler vector** $\boldsymbol{v}_2$, whose signs partition G into two communities